

Predicting the 2025 NFL Regular Season with STAN and Random Forest

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Data Sports Group (2021)

Abstract (Grace)

In this project, we predicted 2025 regular season wins for the National Football League using a Bayesian approach in the analytic software STAN. Bayesian approaches are frequently used in sports forecasting since they encode prior beliefs and uncertainty estimates (Santos-Fernandez et al., 2019). STAN is advantageous for implementing Markov chain Monte Carlo (MCMC) algorithms since it uses the physics-based Hamiltonian Monte-Carlo algorithm, enabling efficient sampling from complex models. We modeled EPA difference as a predictor of game outcome in STAN, using several variants of hierarchical models that represent team ability effects. MCMC diagnostics were found to be favorable for informed-prior and non-weighted variants of the model, and we further used Random Forest classification methods on posterior predictive draws to incorporate additional match information and refine prediction thresholds.

The nuanced forecasting by the Random Forest classifier improved predictive capabilities, with the baseline hierarchical model yielding the most consistent results.

1. Problem Statement (Jimin)

We explored predicting NFL regular-season games using a hierarchical Bayesian linear model in STAN. STAN is an effective choice because it allows us to estimate team-level strengths and quantify uncertainty in game predictions. Our goal is to predict the outcomes of the 2025 NFL regular season by generating posterior predictive samples from a model fit to the games that have already been played by week 13. We also examine several extensions to the model, such as using past-season data to derive informative priors.

2. Data Description (Grace)

2.1 Data Source

The data used in this project came from the “nflverse” ecosystem, which provides publicly available data for NFL analytics and can be loaded efficiently in R with the “nflreadr” package (Ho & Carl, 2025). For our project, we focused on the “schedules” and “team_stats” datasets. “Schedules” provides game-level information on home and away teams, game week, outcomes, and match conditions for completed and upcoming games. “Team_stats” provides information on each team’s performance in completed games, including measurements of EPA. Notably, this dataset includes two rows of data for each game as it provides separate gameplay statistics for both teams. We denote the team being assessed in each row as the “POV” team. Based on our checks, neither of these datasets contained unexpected missing values, duplicates, or outliers, so all entries were kept.

We loaded both the 2024 and 2025 season data, using the 2025 data for prediction purposes and the 2024 season data to inform assumptions. Since the 2025 information updates continuously as games are played, we used snapshots to ensure consistency and reproducibility. Our snapshots included results for games up until and including week 13 of the regular season. In this snapshot, the “team_stats” dataset contained $N = 388$ rows, representing the 194 played games, and 102 columns of statistics for $K = 32$ teams. The corresponding “schedules” dataset had 272 rows and 45 columns and included NA entries for unplayed games. This snapshot

provided an approximately 70%-30% split between past and future games, supplying a reasonable amount of data to fit models while leaving a substantial portion for prediction.

2.2 Data Manipulation

Notable additions that we made to these datasets include computing a general EPA measure for each team and opponent team by adding the passing EPA and rushing EPA for that team as well as assigning integer identifiers to each team and game. We also assigned a recency weight to each game using an exponential decay function that reduces the weight of games earlier in the season. To facilitate analysis, we joined the “team_stats” and “schedules” datasets on both home and away teams, augmenting the “team_stats” structure with outcome information from “schedules”. We built a dataset for future games by extracting rows with NA values in the results column of “schedules”, which indicates unplayed games, and duplicated entries to match the structure of the past-games dataset. After construction, the future-games dataset included $M = 156$ rows, corresponding to the 78 unplayed games. Finally, we constructed a one-hot predictor matrix, which encodes opponent teams for each game-level entry, as well as a vector team-outcome mapping, which matches “POV” team identifiers to games, separately for the past and future games.

3. Methods

3.1 Model Justification (Brady)

We used expected points added (EPA) as our model parameter because it quantifies how much a given play is worth. EPA is calculated as the difference between a team’s expected points (EP) before and after a play. Offensive EP depends on down, distance, field position, time remaining, and score differential. Using a Bayesian approach, we updated a team’s EPA each week to generate a total score reflecting how well a team consistently puts itself in scoring situations.

As a result, this approach helps predict performance because big plays can dramatically shift game outcomes, and EPA also adjusts for factors like garbage-time production. Furthermore, it captures both high-impact plays and the overall body of work from prior games, allowing us to quantify team quality despite random elements such as red-zone luck, tipped-pass interceptions, or missed field goals in close games. In the nflverse package, we used each team’s

passing and rushing EPA, where passing EPA measures value generated on passing attempts and rushing EPA measures value from runs, scrambles, or kneels. These were combined into a total offensive EPA. We then tested the correlation between EPA difference and game outcomes using the 2024 season (Figure 3.1.a). We found that in over 86% of games, the team with the higher EPA wins the matchup, and larger EPA differences are also correlated with a greater margin of victory.

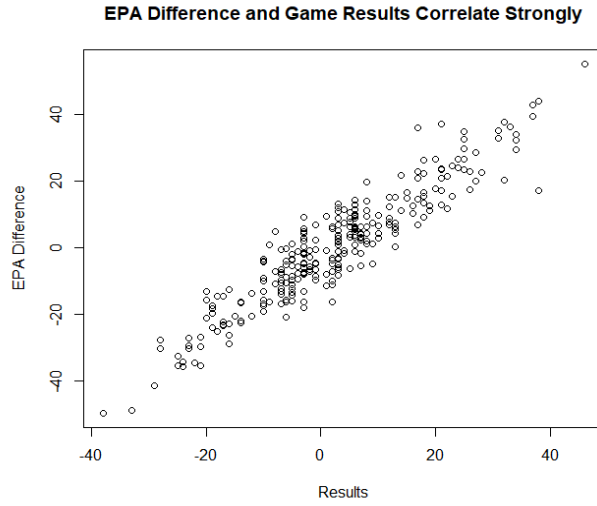


Figure 3.1.a. Scatter plot of EPA difference and game results

3.2 Model Specifications

Basic hierarchical model (Grace)

We modeled EPA as a combination of “POV” team and opponent team abilities. We included different variations of a basic hierarchical model to assess differences in convergence and predictive capacities. In the basic hierarchical structure, we assumed that the ability of individual teams are generated from a shared population distribution with each team having its own deviation. Assuming that we have $g = 1, \dots, N$ games played by $t = 1, \dots, K$ teams, $t(g)$ is the team-outcome mapping vector, and X is the one-hot encoding matrix for the played games, the EPA difference for a given game g can be modeled as:

$$Y_g \sim \mathcal{N}(a_{t(g)} + X_g b_k, \sigma)$$

$$a_t = a_{\text{field}} + a_{t, \text{diff}}$$

$$a_{t, \text{diff}} \sim \mathcal{N}(0, \tau_a)$$

$$b_k \sim \mathcal{N}(0, \tau_b) \quad k = 1, \dots, K, \text{ indexing opponents}$$

This model generally follows the structure of the hierarchical model described in Matsuura (2022). We omitted the b_{field} term described by Matsuura since it would, in our case, encode redundant information. Random intercepts in the a term and fixed effects with a shrinkage prior in the b term capture and constrain team-level differences. We selected weakly informative half-Cauchy(0, 5) priors on the τ_a and τ_b hyperparameters governing standard deviation. Following the specifications in Matsuura (2022) we left the field intercept with a flat prior. Upon inspection, the distribution of EPA difference from the completed games showed multimodality (Figure 3.2.a), suggesting that a mixture model might capture the data more exactly. However, as multimodal sampling in STAN may be unstable and since our goal was to generate predictions rather than precisely reproducing the distribution, we retained the hierarchical structure with a Gaussian likelihood.

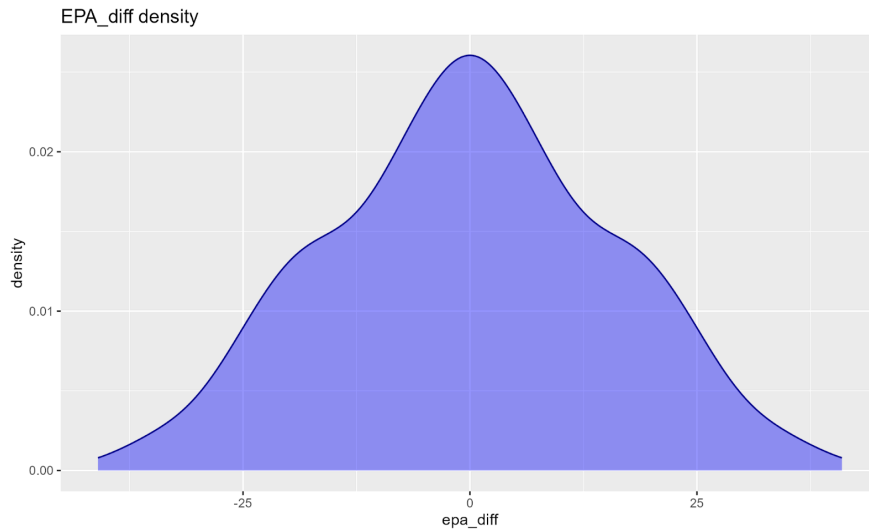


Figure 3.2.a. Density of EPA difference.

Hierarchical model with informed priors (Grace)

Since it was reasonable to assume that hyperparameter values would be relatively stable between successive game seasons, we additionally implemented a model in which τ_a and τ_b were informed by 2024 season performance distributions. We acquired the informed priors by fitting the same hierarchical model to the 2024 season data and extracting posterior summaries for τ_a and τ_b . This method yielded the informed priors of half-Cauchy(7.01, 1.12) for τ_a and half-Cauchy(6.99, 1.12) for τ_b . Since these hyperparameters govern the variability of team intercepts and opponent effects, more informative priors may improve MCMC performance.

They may also improve prediction accuracy as the current-season data has limited sample size and thus potentially limited information.

Recency weighted model (Brady)

When building our model we applied a recency weight that is a three week half life to decay the value of a team's total EPA. This feature allows us to account for roster changes and injuries while still recognizing the team's performance in less recent games. A recency weight also captures the “evolution” of teams throughout the season because an integral part to football is improving week to week and maintaining team morale.

3.3. Classification (Jimin)

We previously showed that EPA difference is strongly correlated with game outcomes (Section 3.1), but this correlation does not fully characterize how EPA difference maps to winning. To study this relationship, we trained a baseline classifier that uses a deterministic threshold and a univariate random forest model. Posterior predictive EPA draws for past games were used to assess model performance, and draws for future games were used to predict the outcomes of the 2025 NFL regular season. After combining home- and away-team posterior predictive values into a home-team perspective (Appendix B), we split the data into training and testing sets using an 80–20 split and evaluated models under four scenarios: (1) true EPA difference in the training set, (2) true EPA difference in the test set, (3) posterior predictive EPA draws in the training set, and (4) posterior predictive EPA draws in the test set.

Baseline Model

The baseline model applies a deterministic threshold to classify the winner of the games from the predicted EPA difference. We tested threshold values from -10 to 10 using the mean 0-1 loss as the evaluation criterion on the training data (Figure 3.3.a). For the basic hierarchical model, we selected -3 as the optimal threshold, while for the hierarchical model with informed priors, we selected -4. They both achieved a mean 0-1 loss of 0.2451613, compared to 0.2709677 when using the threshold of zero.

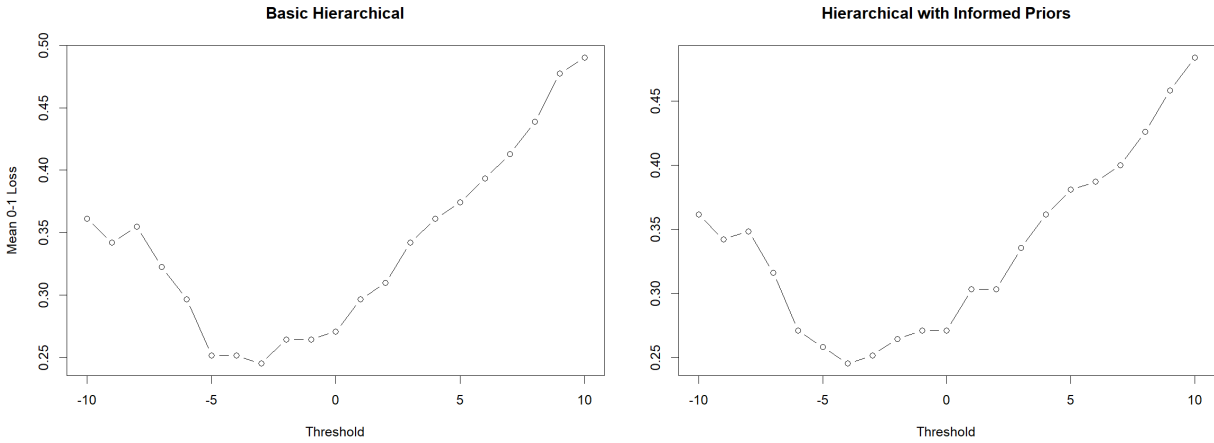


Figure 3.3.a Mean 0-1 loss across threshold values for the baseline model

Random Forest Model

The random forest model is used to translate EPA differences into predicted game outcomes in a univariate setting. We use the random forest as a comparison to the baseline model because it learns a nonlinear probability function rather than relying on a fixed threshold.

4. Results

4.1 MCMC (Grace)

We fit each Bayesian model using STAN in R (version 4.4.3) with the RStan package (version 2.32.7). We set a fixed seed and sampled four chains with 2000 iterations each, including 1000 burn-in iterations. Posterior predictive draws included draws for EPA difference in both past games and future games.

Figure 4.1.a. visualizes aggregated $\hat{R}$ and Effective Sample Size (ESS) metrics across hyperparameters as general convergence trends. The basic hierarchical model and informed hierarchical model saw low $\hat{R}$ values under 1.01 and high ESS values surpassing 50% of iteration count (2000) in most cases. This indicates satisfactory convergence and meaningful posterior estimates. The model variants incorporating recency demonstrated poorer mixing with lower ESS and $\hat{R}$ values above 1.01. This is likely because the decay weights used by the recency model artificially shifts the distribution of the data, making it difficult for population-level standard deviation parameters to converge. Older games also contributed a

lower proportion of information, possibly directly affecting the ESS measures. Additional convergence diagnostics are provided in Appendix A.

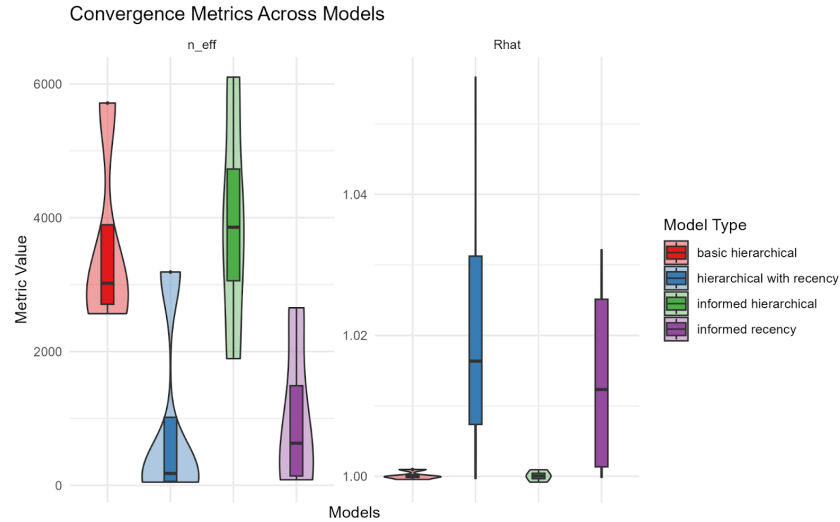


Figure 4.1.a. Violin plots of aggregated convergence metrics.

The posterior mean of the team-level parameters $a_{t, \text{diff}}$ and b_k displayed greater shrinkage towards 0 in recency models compared to the basic hierarchical models (Figure 4.1.b). This indicates that recency models viewed team abilities as being more uniform across the league, posing a potential issue as team abilities are less differentiable. The recency-weighted models also had the shortest intervals, suggesting that more recent games provided consistent information on team ability. Introduction of informed priors had no significant effect on the interval length of hierarchical models but produced noticeably wider intervals for the recency models, indicating that prior-season data may disagree with current-season information when games are weighted. Despite this, the 95% credible intervals for all models were overlapping, demonstrating a somewhat consistent evaluation of team abilities across models.

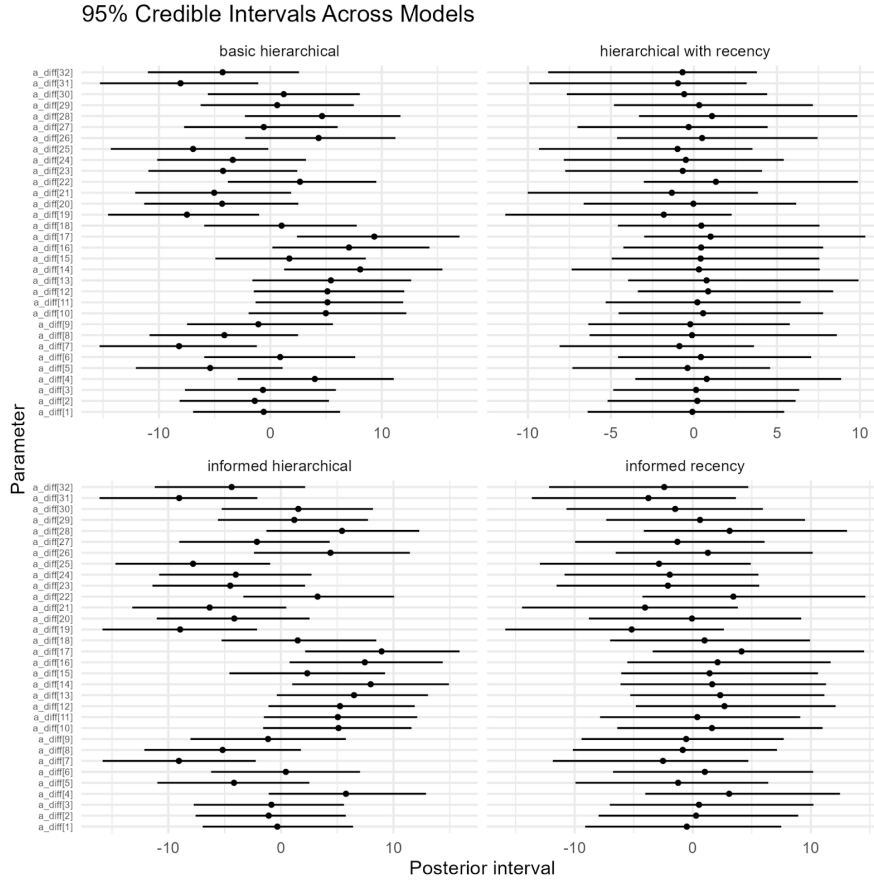


Figure 4.1.b. 95% intervals across models for a_{diff} . The trends for b generally mirror that of a_{diff} .

Since STAN provided posterior draws for past games, we were able to check the predictive accuracy of the models by comparing the generated EPA quantities with actual EPA differences in past games. As expected, the model structure with a normal likelihood was unable to fully replicate the multimodalism of the true EPA difference but generally captured the mean and spread (Figure 4.1.c). The recency model once again exhibited the closest clustering to the mean and relatively lower spread. In contrast, the informed hierarchical model had the lowest concentration of density at the mean. Finally, we obtained root mean squared error (RMSE) and mean absolute error (MAE) metrics, shown in Figure 4.1.e. The informed models, in general, yielded lower error measurements than their non-informed counterparts. For instance, the informed hierarchical model achieved RMSE around 12.11 compared to 12.65 for the non-informed. The basic hierarchical models also produced lower error than the recency-weighted variants. Since both convergence and error metrics deemed the

recency-weighted models less effective, they were not included in the following classification analyses. Additionally, we compared the error values to error measurements obtained by a null model that predicts a mean EPA difference of 0 for all games. All models performed better than this baseline, confirming that they have notable predictive power despite the Gaussian likelihood being only an approximation of the true distribution.

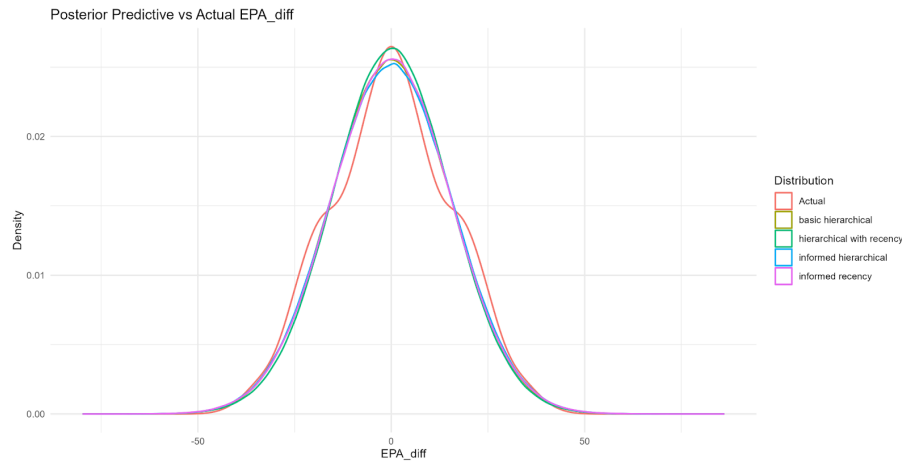


Figure 4.1.c. Posterior draw densities compared to actual EPA difference distribution.

model	rmse	mae
basic hierarchical	12.65098	10.114882
hierarchical with recency	14.61468	11.683703
informed recency	13.70394	10.935535
informed hierarchical	12.10844	9.673944
null baseline	15.54838	12.475067

Figure 4.1.d. RMSE and MAE metrics.

4.2 Classification (Jimin)

Baseline Model

I. Mean 0-1 Error

We measured the mean 0-1 loss using the best thresholds for each hierarchical model in four scenarios (Table 4.2.a). The positive class denotes that the home team is predicted to win. The error is the lowest when the true EPA differences are supplied as the input, and it increases

when posterior predictive EPA differences are used. The posterior predictive EPA differences for the test set produce the highest error, which reflects the compounding uncertainty from both estimation and out-of-sample prediction.

	Basic Hierarchical		Hierarchical with Informed Priors	
	True EPA_diff	Predicted EPA_diff	True EPA_diff	Predicted EPA_diff
Training	0.148387096774194	0.245161290322581	0.141935483870968	0.245161290322581
Testing	0.153846153846154	0.333333333333333	0.128205128205128	0.384615384615385

Table 4.2.a Mean 0-1 error using the best thresholds

II. Confusion Matrix

We plotted confusion matrices for the two hierarchical models, using the posterior predictive EPA differences for the test set and applying the respective best thresholds (Figure 4.2.a). We noticed that the true positive rate and false negative rate remain constant across the hierarchical models, but the true negative rate decreases and the false positive rate increases for the hierarchical model with informed priors. This suggests that using informed priors produces greater variability near the decision boundary, particularly on the negative side of the distribution, which leads the deterministic threshold to classify more losses as wins.

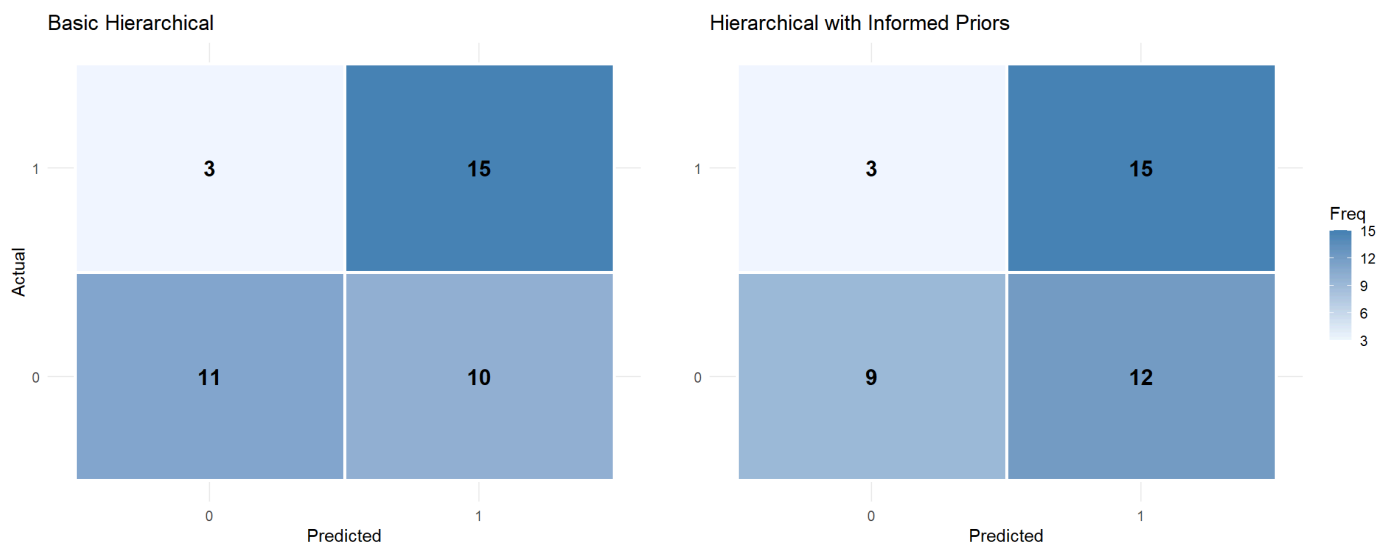


Figure 4.2.a Confusion matrices for the baseline model

III. Density Plot

We plotted density curves of the predicted EPA differences for the combined training and test dataset, separated by the true game outcome (home win vs. home loss), and included a vertical dashed line indicating each model's optimal threshold (Figure 4.2.b). The two plots show very similar density curves in both shape and magnitude, suggesting that the greater false-positive error for the hierarchical model with informed priors arises not from differences in the underlying EPA distribution, but from its optimal threshold being one unit farther to the left than that of the basic hierarchical model. As mentioned earlier, the training dataset was used to identify these thresholds. Because the optimal thresholds differ only marginally, it is likely that the models diverge on just a small subset of training observations near the decision boundary.

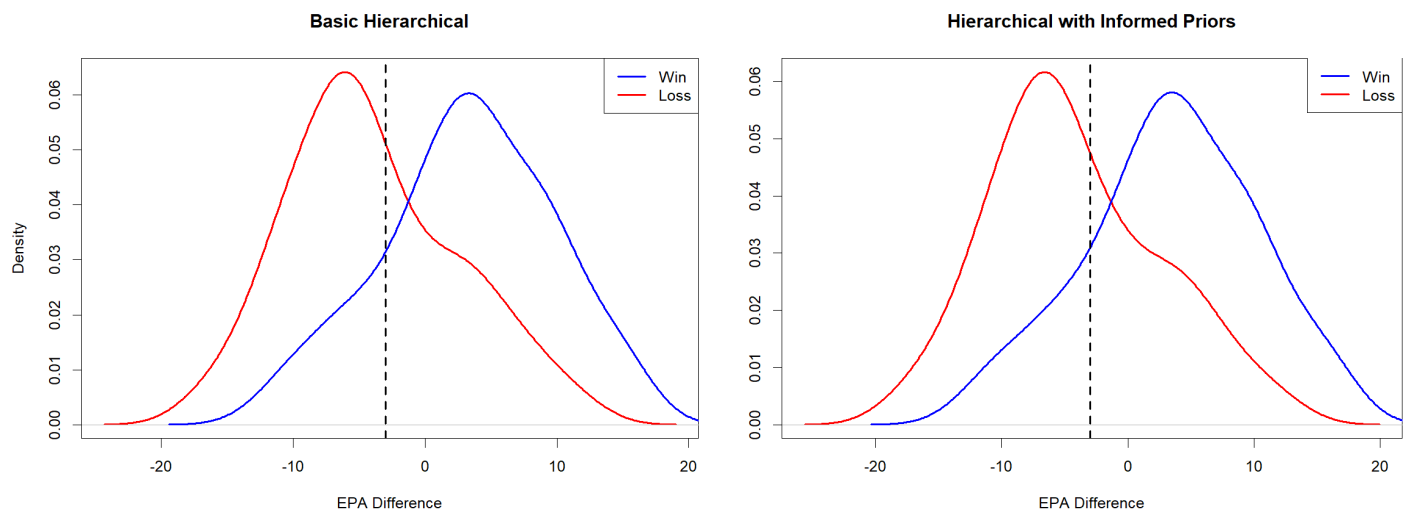


Figure 4.2.b Density plots of EPA differences with best thresholds

Univariate Random Forest Model

I. Mean 0-1 Error

We measured the mean 0–1 loss for the same four scenarios as in the baseline model (Table 4.2.b). The errors for the hierarchical models were identical in all scenarios except when the posterior predictive draws for the training set were given. In this case, the hierarchical model with informed priors achieved a lower error, consistent with its lower RMSE as stated in Section 4.1. This suggests that the informed priors improved the model's accuracy of posterior predictive EPA differences on the observed games, and that this improvement translated into better classification performance when using those predictive draws. Compared to the deterministic threshold method, the univariate random forest model achieves lower error in all four scenarios,

indicating that the random forest better captures the relationship between EPA differences and game outcomes.

	Basic Hierarchical		Hierarchical with Informed Priors	
	True EPA_diff	Predicted EPA_diff	True EPA_diff	Predicted EPA_diff
Training	0.1225806	0.2645161	0.1225806	0.2580645
Testing	0.1025641	0.2820512	0.1025641	0.2820513

Table 4.2.b Mean 0-1 error for the univariate RF model

II. Confusion Matrix

We plotted confusion matrices for predictions generated by the two hierarchical models, using the posterior predictive EPA differences for the test set as inputs (Figure 4.2.c). In addition to the same mean 0-1 error of 0.2820513, their confusion matrices were also identical. Compared to the baseline model, the univariate random forest classifier achieves a noticeably better true-negative and false-positive rate, indicating that it is better at separating losses from wins. This improvement reflects the fact that the random forest learns a nonlinear probability function that captures uncertainty in outcomes near the decision boundary.

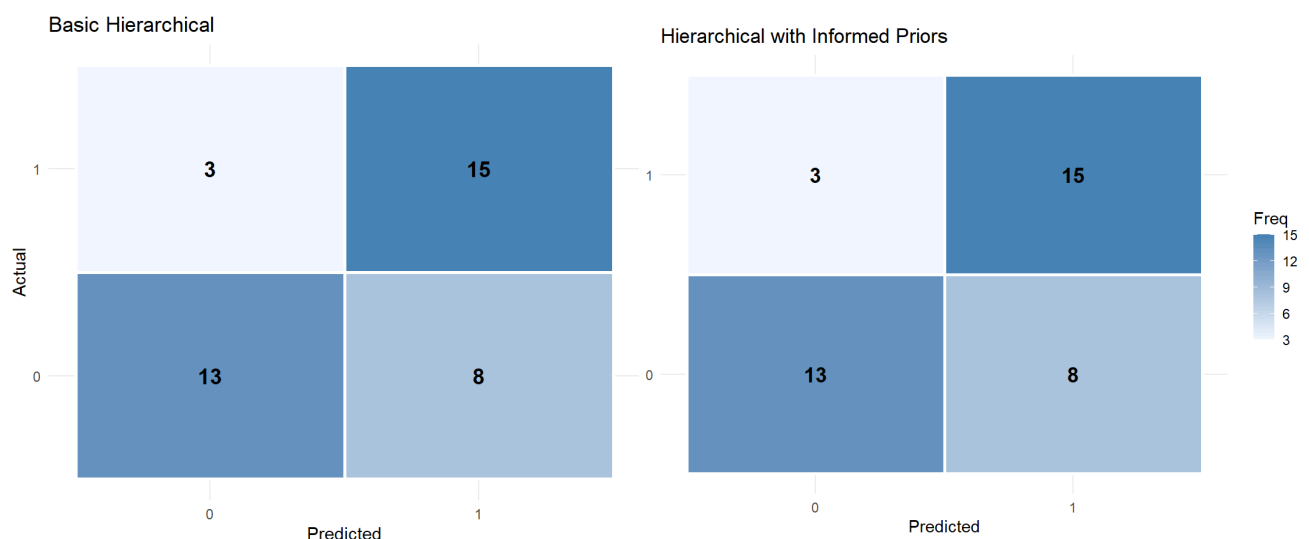


Figure 4.2.c Confusion matrices for the univariate RF model

III. Decision Threshold Plot

We plotted decision-threshold curves to examine the behavior of the random forest model (Figure 4.2.d). These plots show the predicted probability of being classified as the positive class as a function of EPA difference. For both the basic hierarchical model and the hierarchical model with informed priors, the point at which the dashed horizontal line at probability 0.5 intersects the curve lies slightly to the left of zero, which is consistent with the optimal thresholds identified earlier in the baseline analysis. Despite having similar effective decision boundaries, the random forest model outperforms the baseline threshold approach. This suggests that the random forest better captures the uncertainty in game outcomes near the decision boundary. For example, around an EPA difference of zero, the predicted probabilities span a wide range, reflecting that games with small EPA differences may plausibly be wins or losses. A deterministic threshold cannot represent this variability.

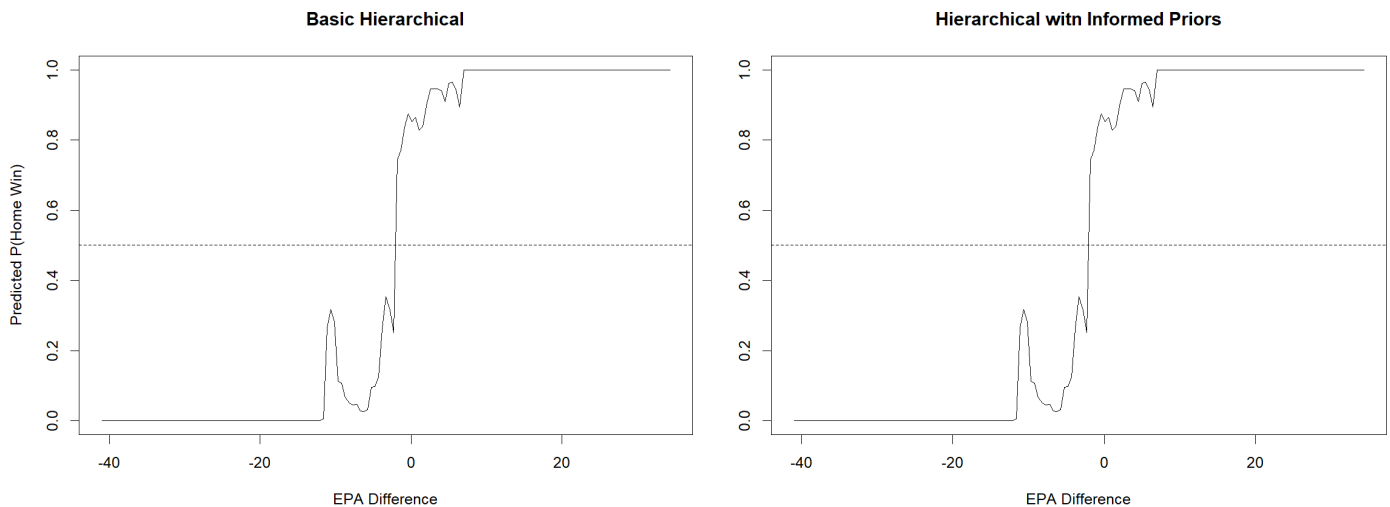


Figure 4.2.d Decision boundary plots for the univariate RF model

4.3 Predictions (Jimin)

We used the models to generate season predictions, focusing on the top-performing teams in the AFC and NFC and the three lowest-performing teams overall. The results were largely consistent, except for the baseline model when informed priors were used. As noted earlier, using informed priors for the baseline model introduced greater variability near the decision boundary, which helps explain why the corresponding baseline predictions diverge from those produced by the other methods.

	Baseline Model		Univariate RF	
(Hierarchical Model)	Basic	Informed	Basic	Informed
AFC	NE	NE	NE, DEN	NE, DEN
NFC	LA	SEA	LA	LA

Table 4.3.a Predicted top performing teams in AFC and NFC

	Baseline Model		Univariate RF	
(Hierarchical Model)	Basic	Informed	Basic	Informed
Worst (1st)	TEN	TEN	TEN	TEN
2nd-Worst	LV	LV	LV	LV
3rd-Worst	WAS, NYJ	WAS, NYJ, CIN	WAS, NYJ	WAS, NYJ

Table 4.3.b Predicted three lowest-performing teams

4.4 Extensions on Random Forest Classification (Brady)

Random forest models are designed to take advantage of multiple features because they can capture nonlinear relationships and interactions across predictors. Although our univariate random forest model uses only a single input (EPA difference) due to limited data availability, this setup does not fully leverage the strengths of random forests. Therefore, we extend the model to a multivariate version by incorporating additional covariates. This is not a robust Bayesian model as we did not run MCMC chains for each new in-game prediction feature and more of a test to see what could be added alongside EPA in future work.

I. Features

For our multivariate random forest model we added the eight in game prediction features of turnover margin, penalties, field position, field goal percentage, field goal percentage beyond 50 yards, sack margin (difference between defensive sacks and sacks taken), total first downs, and passing completions percentage above expected (CPOE) that capture elements of the game that EPA fails to do. We also included four pregame features which are roof, surface, location, and days of rest. These features were added to see if there were known factors prior to the game that we could use to improve the model without adding additional predictions. We fit a random forest with these features for both our basic and informed hierarchical models.

II. Feature Importance

When testing the importance of all features included in the random forest (Figure 4.4.b). Both hierarchical models had the four most important features as field position, posterior predictive EPA differences (pred), turnover margin, and CPOE, while the pregame known features were not very impactful to modeling accuracy. While it would be difficult to run MCMC chains on all the added features, it would be reasonable for us to look into adding a couple of these features to our model. The known features do not appear effective enough to implement.

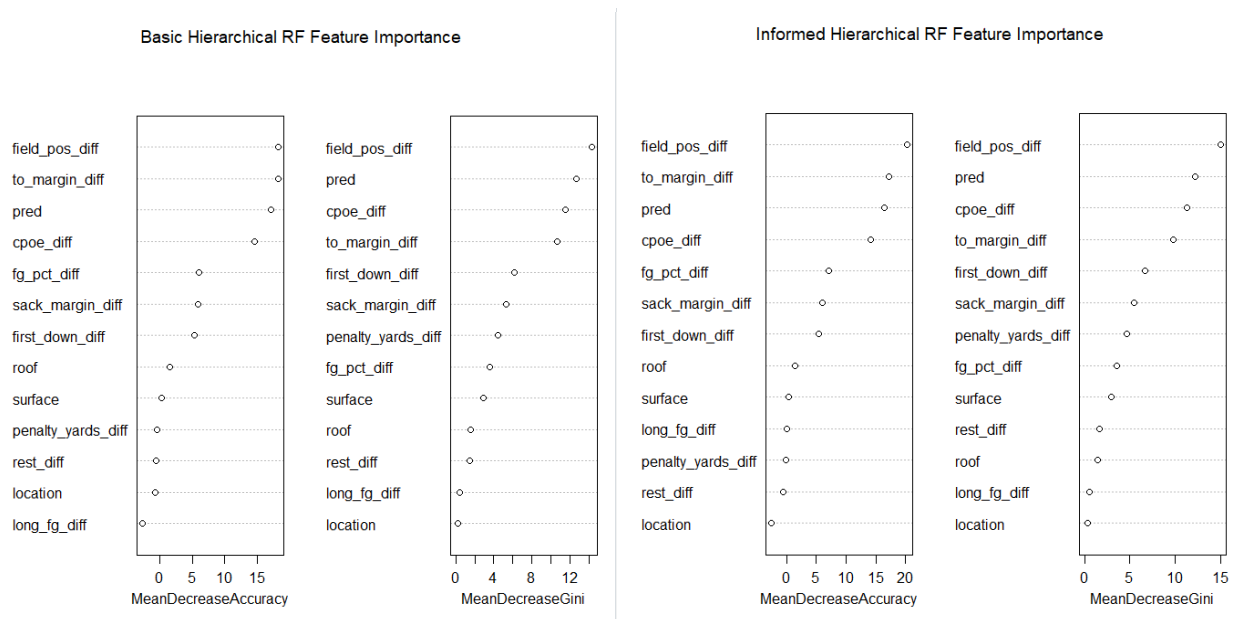


Table 4.4.b Multivariate Random Forest Feature Importance

III. Confusion Matrix:

We plotted confusion matrices for the two hierarchical models using the random forest (4.4.c). The basic hierarchical model had an error rate of around 23% and the informed hierarchical model had a slightly lower error rate around 18%, which are both noticeable improvements from other models meaning that the new features would be able to improve prediction accuracy if implemented. The trend of overpredicting false positives continues, which suggests there may still be an element outside of EPA and the additional features that we are failing to capture.

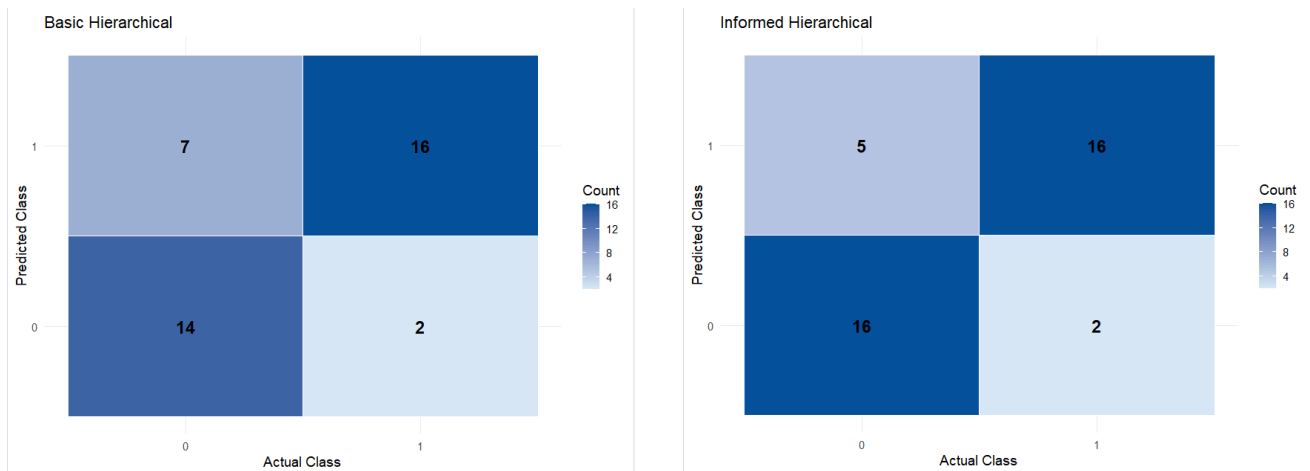


Table 4.4.c Confusion Matrix for Multivariate RF Model

5. Limitations and Future Work (Brady)

5.1 Limitations

Many of the limitations we faced were due to the wide variety of factors that create a winning football team and the resources we had to account for them. The best example of this is a team's injuries. Injuries are common due to the violent nature of football and are difficult to account for without a live injury report. It is also challenging to quantify how much injuries affect performance because a player's success depends on the performance of their teammates. While we applied a recency weight to approximate this, it is only a constraint and does not reflect who will actually contribute in the following week.

Another issue was the limited data in a 17-game season. Because rosters can change drastically from year to year, the most reliable information comes from the current season alone. The main problem this presents is the limited number of common opponents teams have faced to

that point. Although variation in opponent caliber may be similar, predicting how an offense will perform against a new defense remains uncertain, as a defense may have a strength that exploits an offensive weakness and vice versa.

One more limitation was the variability in how a team performs week to week. Minor changes in scheme or playcalling can cause a team to outperform previous weeks, while factors like locker-room tension or a devastating loss can reduce future production. Lastly, we were on a computational power constraint that prevented us from pursuing some of the more computationally intensive ideas we had for improving the model.

5.2 Future Work

In future work we thought that we could improve our hierarchical model by modeling the priors regression intercepts (team-level strengths) as joint distribution, allowing teams strengths to be correlated. This would improve our model because of the constant evolution in schemes, playstyles, and rules that affect the value of passing, rushing, pace, and defensive ability. Recent rules changes such as how roughing the passer and pass interference is officiated has created more value to passing the ball due to the chance an offense may get free yardage on the plays that would have previously resulted in an incomplete pass or sack. Another element of our model that could be further looked into is applying the features tested in section 4.4 because it seems that features like field position and CPOE will be able to apply context to our model that EPA fails to capture.

References

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Appendix

Appendix A: Additional Convergence Diagnostics (Grace)

Figure A.1. shows the trace plots for all hyperparameters, reflecting general convergence patterns. Trace plots show consistent convergence for the informed hierarchical and basic hierarchical models across all four chains. Convergence is the poorest for the hierarchical recency model in the τ_a and τ_b parameters, and is slightly improved in those parameters when informed priors are incorporated. This is reasonable since informed priors act directly on those parameters, pulling them towards a known previous season value. Convergence for the other parameters appears to be relatively satisfactory across models, although the recency models converge to higher values of σ in general than the non-weighted models. This is likely because the non-weighted models have a greater effective sample of game observations since the recency weighting essentially reduces the value of earlier games as observations.

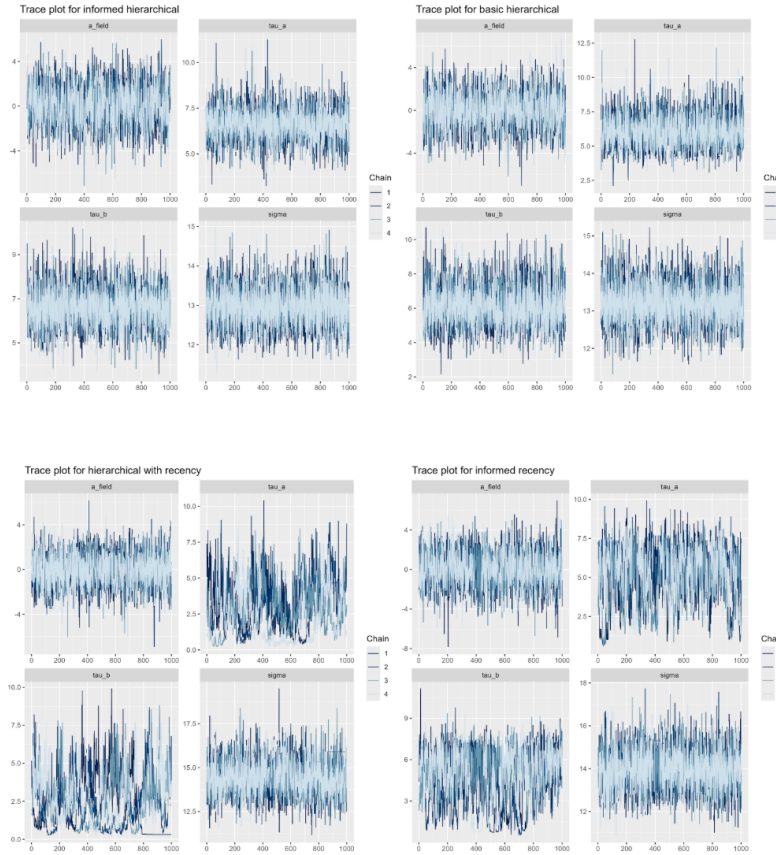


Figure A.1. Trace plots for hyperparameters.

Figure A.2. displays posterior density overlays for hyperparameters as well as one team's team-level parameters. These plots show the density curves for each parameter across the four chains. These generally indicate similar patterns of convergence as observed with the trace plots, with inconsistent densities in the informed recency and hierarchical recency models for the τ parameters. These plots more intuitively display how the informed priors on these parameters pull their value towards a set mean in the informed recency model.

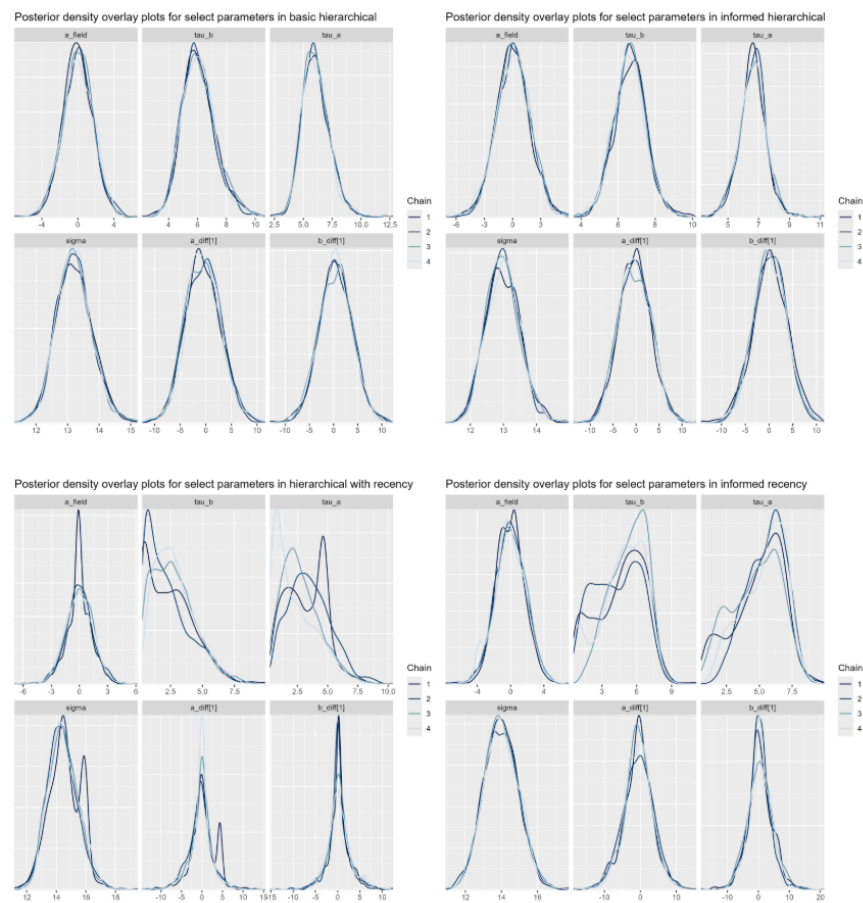


Figure A.2. Density overlay plots for select parameters.

Figure A.3 shows a bar plot representation of convergence metrics across models, providing another visualization of convergence patterns. Once again, the τ_a and τ_b hyperparameters demonstrated the poorest ESS for the recency model variants, reflecting the fact that these models have fewer meaningful observations to form estimates due to the

decay-function weighting. a_{field} estimates have lower ESS in the informed models, likely also a consequence of the fact that this hyperparameter is not given an explicit prior in STAN sampling and thus prior season patterns are not captured. Incorporating a prior for this hyperparameter would likely improve performance.

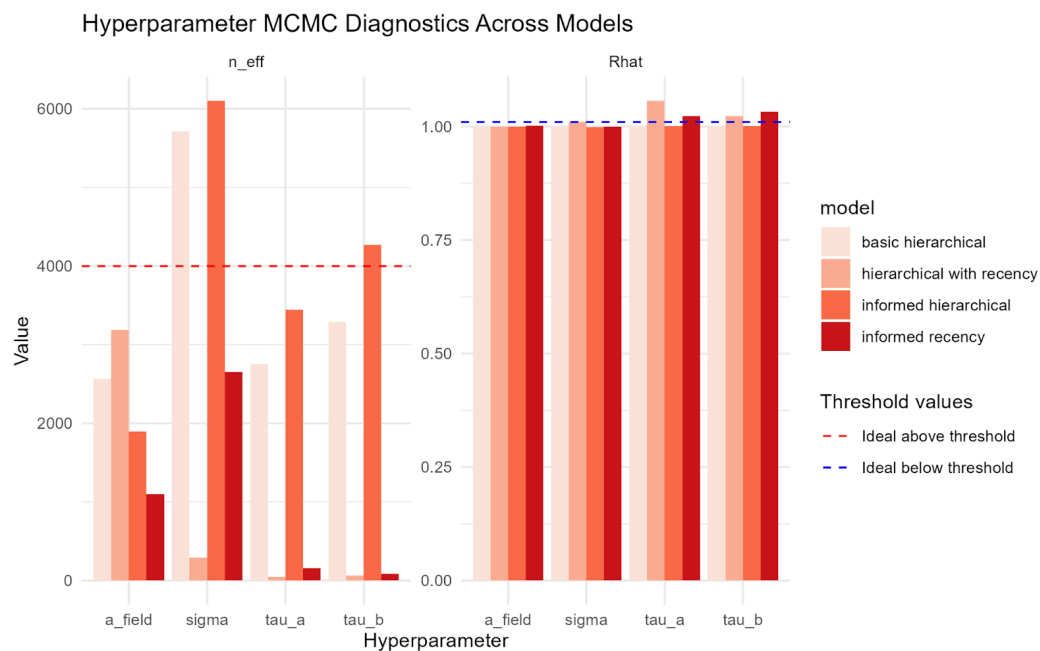


Figure A.3. Barplot representation of convergence patterns.

Appendix B: Data Preparation for Classification (Jimin)

In the combined dataset for past and future games, each game is represented by two rows: one from the home team's perspective and one from the away team's perspective. Because the hierarchical model uses a POV-specific team intercept and opponent indicator, the posterior predictive samples for the same game do not always align, although they do agree in most cases. For example, the root mean squared error (RMSE) of the posterior predictive draws of EPA difference for past games is 12.18612 when the "POV" team is the home team and 12.18845 when the "POV" team is the away team.

To reconcile the two predictions produced for each game, we first separated the posterior predictive samples by whether the "POV" team was the home or away team and then considered two approaches for retaining only one row per game, with the home team defined as the "POV" team:

1. Average the two predicted EPA differences (after converting the away-POV prediction to the home team's perspective).
2. Select the predicted EPA difference with the greater magnitude.

To formalize the first method, let $EPADiff_{home}$ denote the predicted EPA difference when the "POV" team is the home team and $EPADiff_{away}$ the predicted EPA difference when the "POV" team is the away team. Converting the away-POV prediction to the home perspective yields $-EPADiff_{away}$, so the averaged prediction is $(EPADiff_{home} - EPADiff_{away})/2$.

The RMSE of the first approach is 12.18626 whereas selecting the value with the greater magnitude yields a lower RMSE of 12.16927. This difference may arise because averaging pulls the predicted EPA differences toward zero when the two predictions disagree, while using the greater-magnitude prediction preserves the model's implied directional signal for the game. Hence, we adopted the second approach to resolve the two predictions.

Appendix C: Additional MVRP (Brady)

I. Feature Selection:

Turnover margin was added because of its strong correlation to win percentage. Penalties were added because they are factored into EPA; however, defensive penalties are often the fault of the defense and not a result of a powerful offense. Field position was added because Expected

Points is a function of down, distance, and total yards until the endzone, due to a team that starts in advantageous field position, thus resulting in fewer “potential points” available. We also added field goal percentage and field goal percentage beyond 50 yards because this is another element that should affect EPA, but is not accounted for, because if a team has a more efficient or stronger kicker, then they will be able to score more points without needing to get closer to scoring. Sack margin and total first downs were added to account for the playmaking abilities of a team because ultimately, these are two key factors in keeping a drive going on offense and stopping a drive on defense. Lastly, CPOE was added to factor in quarterback play; although football is a team sport, the power of an offense is largely contingent on quarterback play. We also included four pregame features, which are roof, surface, location, and the difference in rest between the two teams. These features were added to see if there were known factors prior to the game that we could use to improve the model without adding additional predictions. We fit a random forest with these features for both our basic and informed hierarchical models.

II. ROC Plots:

Figure C.1 shows the ROC curves from the multivariate random forest. Both models show excellent classification ability, achieving a true positive rate quickly, with the basic model showing slightly more fluctuation.

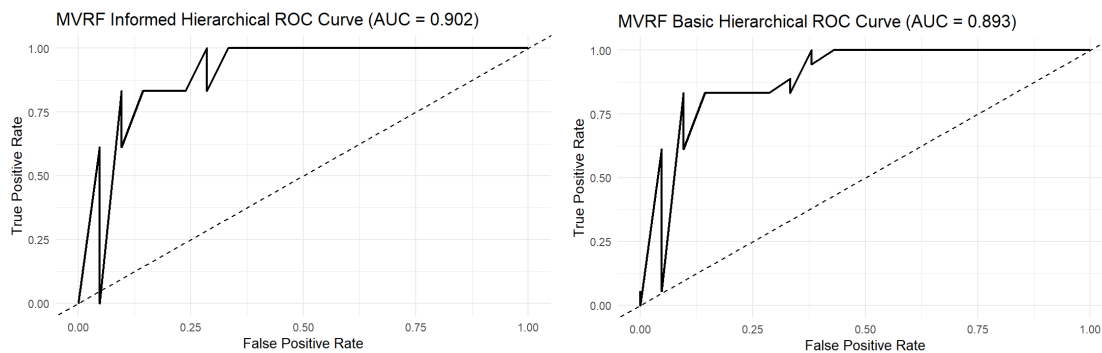


Figure C.1. MVRF ROC Curve

III. OOB plots

Figure C.2 shows the multivariate random forest OOB plots. The information model converges faster than the basic model while the error rates seem to be consistent between the two. The error rate for predicting losses is again higher in both models.

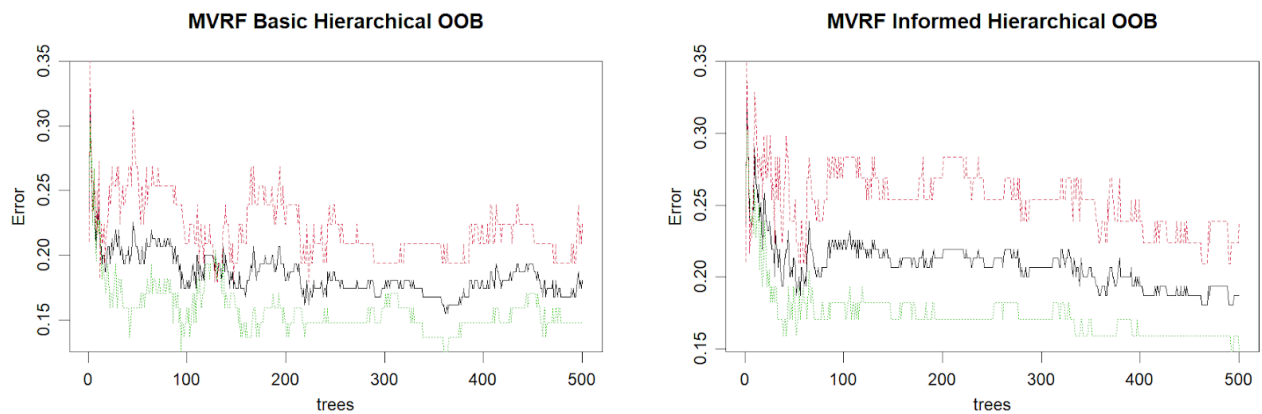


Figure C.2. MVRF ROC Curve